SAI TEJA SRIVILLIBHUTTURU

saiteja.srivilli@gmail.com   +1 (682) 234-3567   github.com/saitejasrivilli   linkedin.com/in/saitejasrivilli

Skills

ML Models & Algorithms: Python, PyTorch, **supervised learning** (XGBoost, LightGBM, CatBoost, neural networks), **unsupervised learning** (SVD, collaborative filtering), Learning-to-Rank (TF-Ranking), GBDT, optimization methods (LambdaLoss, gradient descent)

Statistical Modeling & Evaluation: Bayesian inference, AUC/LogLoss/Gini metrics, NDCG optimization, model selection, hyperparameter tuning, A/B testing, statistical significance testing

Production ML: Real-time inference (<1ms latency), feature engineering, PySpark pipelines, model evaluation frameworks, AWS (S3, Lambda, SageMaker), Docker, Kubernetes

Professional Experience

University of Texas at Arlington — Aug 2024 – Present
Machine Learning Engineer — Texas, US

- Engineered **large-scale ML** system for information retrieval over 1K+ document corpus, achieving **92% accuracy** with **sub-100ms** query latency on AWS; communicated architecture decisions with 5-person team and presented optimization results to stakeholders.
- Constructed **real-time ML data pipeline** in PyTorch on distributed compute; trained and optimized path prediction models, cutting end-to-end routing latency **35%**; collaborated with infrastructure team to resolve compute bottlenecks.
- Formulated high-throughput **model inference** serving layer with RESTful APIs in Python; scaled to 50K+ daily requests and reduced serving latency **25%**.
- Mentored 2 junior ML engineers on supervised learning model selection, hyperparameter tuning, and statistical evaluation; provided code reviews and guided model deployment best practices.

DentalScan — Dec 2025 – Feb 2026
Machine Learning Engineer Intern — Remote, US

- Strengthened AI-driven diagnostic workflows by implementing policy-based guardrails, structured output validation, and escalation logic using rule engines across 5 diagnostic tasks; reduced unsafe inference exposure **40%** (0.8% -¿ 0.48% false positive rate); briefed 3 clinical teams on safety improvements.
- Programmed automated evaluation pipelines with metric-based promotion gates (accuracy, precision, F1, recall); prevented **30%** of high-risk model regressions from reaching production across 6 failure categories; collaborated with QA team to define regression detection thresholds.
- Accelerated retraining cycles **25%** (14 days -¿ 10.5 days) via AWS S3- and Lambda-based preprocessing pipelines with version-controlled datasets and automated drift detection; documented pipeline architecture for team handoff.

Tata Consultancy Services — Jun 2019 – May 2023
Software Development Engineer – Backend & Infrastructure — Chennai, India

- Led and scaled microservices backend exposing high-throughput RESTful APIs processing millions of records; slashed latency **45%** via MySQL optimization and Redis caching; coordinated with 3 teams on API contract changes.
- Established real-time monitoring service tracking microservice health; cut incident response time **40%** via Prometheus/Grafana dashboards; liaised with oncall team to gather monitoring requirements.
- Automated deployment workflows using CI/CD pipelines; slashed release cycle time **30%** and enforced configuration standards; presented deployment best practices to engineering team.

Projects

Real-Time Bidding (RTB) System: GBDT CTR Predictors — GitHub

- Developed production-grade **CTR prediction** models comparing **XGBoost**, **LightGBM**, and **CatBoost** on synthetic RTB data: **XGBoost AUC 0.5662**, **LightGBM LogLoss 0.2237** (best), **CatBoost 0.5494** – all with **sub-0.001ms inference latency** per sample.
- Spearheaded comprehensive **GBDT** benchmark suite measuring AUC, LogLoss, Gini, training time, and **inference latency**; generated production-ready model comparison enabling real-time serving optimization.
- Orchestrated complete RTB engine with second-price auctions, eCPM **ranking**, token-bucket budget pacing, and **<100ms** latency achieving **2000+ QPS** throughput.

Sponsored Search Ranking: Learning-to-Rank System — GitHub

- Engineered two-stage **ranking** pipeline: FAISS HNSW candidate retrieval (˜3ms) + **TF-Ranking** reranker with **LambdaLoss** for listwise NDCG optimization and IPS position bias correction, improving ranking quality **18%** vs baseline.
- Established PySpark **feature engineering** pipeline with Airflow DAG orchestration for large-scale feature computation; served via FastAPI achieving **<50ms** end-to-end latency on 1M+ queries daily.

Amazon Product Recommender System: SVD Collaborative Filtering — GitHub

- Developed large-scale product recommendation system using **SVD-based collaborative filtering** over 1.7M+ user-item interactions, achieving **0.78 NDCG@10** on holdout test set; exposed **inference** via FastAPI with PostgreSQL backend.
- Created Kafka producer/consumer pipeline to stream real-time user interactions and automated model retraining on new batches, reducing stale recommendations **40%**.

- Launched as production-grade serving endpoint with **sub-100ms** recommendation latency, supporting **10K+ daily active users**; articulated system design tradeoffs in technical design review.

Education

University of Texas at Arlington – Master of Science in Computer Science Aug 2023 – May 2025
Andhra University – Bachelor of Technology in Computer Science Jun 2015 – Apr 2019